

WASP-36b

R-2231

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_140102 · created 2026-08-15T01:13:15+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456659.82974 +/- 0.0009 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.048
Transit depth (Rp/Rs)^2	2.73 +/- 1.58 [%]
Orbital Inclination (inc)	81.28 +/- 2.88
Airmass coefficient 1 (a1)	0.69 +/- 0.4
Airmass coefficient 2 (a2)	0.3 +/- 0.25
Scatter in the residuals of the lightcurve fit is	58.0 %
Best Comparison Star	None
Optimal Aperture	2.88
Optimal Annulus	3.75
Transit Duration (day)	0.056 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.165 ± 0.048 vs 0.1368 (deviation 0.52σ)
Duration fit vs published	0.056 d vs 0.0758 d (deviation 0.57σ)
Mid-time O–C	-0.6 min
Depth SNR	3.0
Residual scatter	58.0 %

Light curve

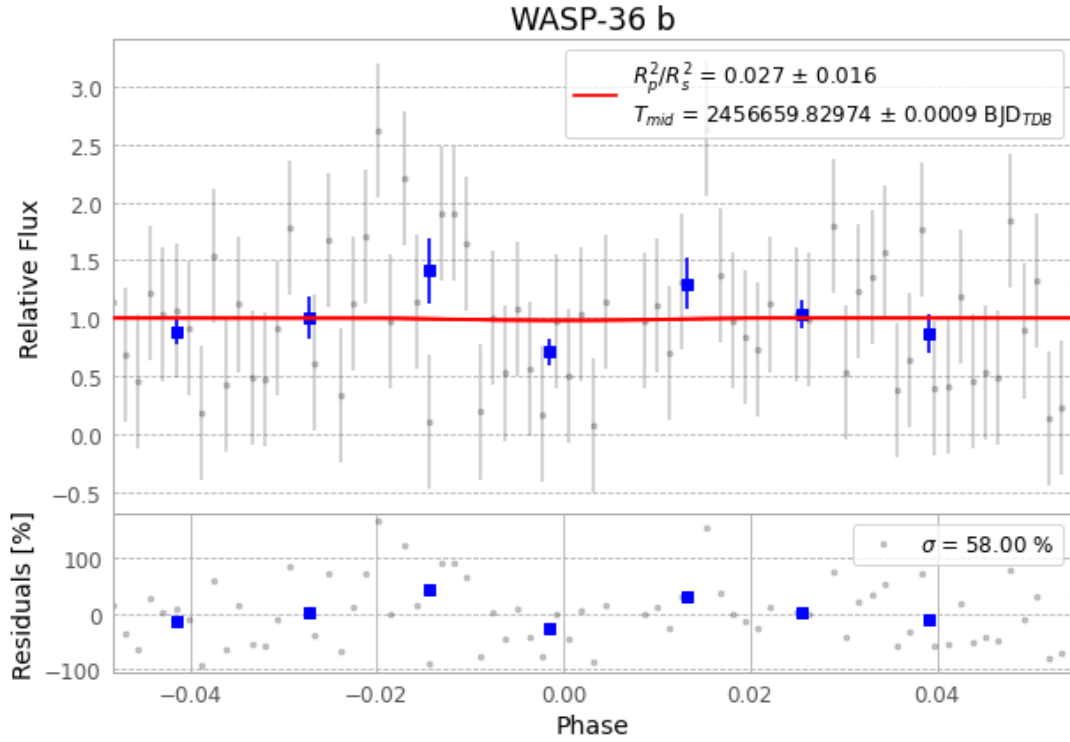


Figure 1 — FinalLightCurve_WASP-36 b_2014-01-01.png (SHA-256 826b8a67f0623b3f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [316, 375], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b7a8380f7ae7fe7e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

77 FITS frames (51 MB), WASP-36140102060012.FITS → WASP-36140102094812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-140102.log (14929286ff9f...), FinalLightCurve_WASP-36 b_2014-01-01.png (826b8a67f062...), FinalLightCurve_WASP-36 b_2014-01-01.csv (3f5f03342af9...)

Compute resources

Wall time 128.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 01:13Z.